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Applications to Enterprise

The conjectures outlined in the last chapter derive their plausibility from everyday experience. As we will see in Part 2, they are also grounded in the ideas of some towering twentieth-century intellects, including John Maynard Keynes and Herbert Simon. But I cannot demonstrate their validity through logico-scientific methods (more on that in Part 4). Instead, I illustrate their utility through two entrepreneurial applications. The applications—analogous to the extended range and comfort of a modern e-bike—are covered in detail in Parts 3 and 4. In this chapter, I preview

- Why entrepreneurship is a suitable venue for my applications.
- The two applications, namely how uncertainty affects the specialization of entrepreneurial initiatives and the nature of entrepreneurial discourse.
- What lies ahead in the next parts of the book.

1. Why Entrepreneurship?

Surging Interest William Baumol's 1968 article in the *American Economic Review* lamented the absence of the entrepreneur from economic theory. The entrepreneur, Baumol wrote, had frequently appeared in the writings of classical economists but as a "shadowy entity without clearly defined form and function." Only Schumpeter and Knight had "infus[ed] him with life" and "assign[ed] to him a specific area of activity to any extent commensurate with his acknowledged importance." Subsequently, the entrepreneur had "virtually disappeared from the theoretical literature."[1]

The theoretical firm, Baumol continued, was now"entrepreneurless—t he Prince of Denmark ha[d] been expunged from the discussion of *Hamlet*." The firm was assumed to "perform a mathematical calculation which yields optimal (i.e., profit maximizing) values for all of its decision variables." There was "no room for enterprise or initiative. The management group becomes a passive calculator that reacts mechanically to changes imposed on it by fortuitous external developments over which it does not exert, and does not even attempt to exert, any influence. One hears of no clever ruses, ingenious schemes, brilliant innovations, of no charisma or of any of the other stuff of which outstanding entrepreneurship is made; one does not hear of them because there is no way in which they can fit into the model."[2]

In my analysis, the developments in microeconomics during and after the 1930s (detailed in chapter 6) that excluded uncertainty from mainstream theory also banished entrepreneurship. Then, starting in the 1990s, business schools went on a hiring spree for new faculty to satisfy surging demand for entrepre-neurship courses.[3] Many of the new faculty hired were capable, ambitious young economists. They secured tenure through prolific publications on entrepreneurial topics in top economics journals. Their success encouraged more such research (as discussed in chapter 13).

Complementary Potential The recruits did not, however, emphasize uncertainty in their research. Mainstream economics had continued down the explicitly uncertainty- free path it started on in the 1930s (reviewed in chapters 6 and 7). As we will see, the popular new specialties in economics focused on concerns about misaligned incentives, not doubts about misjudgments. And while young economists had ample scope for researching entrepreneurial topics, such as venture capital contracts, if they wished to get published in top journals (and thus tenure), they could not stray far from the mainstream. Baumol's Hamlet was now firmly in the kingdom— but without the doubts of Shakespeare's protagonist.

It is hard to imagine entrepreneurial initiatives that do not involve meaningful doubts about one-offs. Including such doubts thus offers a significant opportunity for scholarly inquiry grounded in real-world concerns. At the same time, we do not have to reject mainstream concerns about misaligned incentives. Including the conjectures about doubts and justification described in the last chapter can add to our knowledge of enterprise, coexisting with the research done using mainstream incentive-centric approaches.

2. Specialization and Discourse

Lost Cause Knight's 1921 book distinguished between uncertainty and risk for a reason: to propose that true profit requires bearing (taking "responsibility" for) uncertainty rather than risk. In Knight's theory, providing capital for risks that can be calculated from the laws of probability or statistical tables only earns the going market rate for risk-bearing. (*Conceptually*, according to Knight, the market rate of return must be excluded from true profit, although no one actually does this.) Moreover, responsibility for uncertainty is an entre-preneurial function, and the return for performing this function is the source of an entrepreneur's profit.[4]

Although I find Knight's "no-uncertainty, no-profit" thesis appealing, I see little hope of its acceptance in mainstream economics. As I show in Part 2, eco-nomics has well- established theories of profit that ignore uncertainty. Instead of debating the true nature of entrepreneurial profits, I try to show how my modified version of uncertainty helps explain the specialization of enterprise in the organizations that populate the entrepreneurial ecosystem and the role of entrepreneurial discourse.

Specialization When Knight's book was published in 1921, large, professionally managed firms were transforming the entrepreneurial landscape. Knight presciently saw the rise of large firms as an important development for entre-preneurship. But his as-it-was-happening account did not include historian Alfred Chandler's retrospective raison d'être of large corporations: realizing sig-nificant economies of scale and scope. Furthermore, Knight depicted decisions in large organizations as following a simple vertical order. Employees on lower rungs exercise judgment in routine matters, passing up nonroutine matters to bosses.

But economies of scale and scope demand more than simple hierarchies, as Chandler later documented. Businesses attempting to achieve these economies face severe coordination problems. Organizational pyramids and top- down control cannot ensure the alignment of the many parts and activities. Yet misspecification of small parts, like the O-ring seal in the space shuttle *Challenger*'s rocket booster, can have catastrophic consequences. Large corporations cope, as best they can, by establishing intricate, consultative routines with inputs from specialized staff and multifunctional teams to complement hierarchical control. Strict routines also protect stockholders from managerial carelessness. This helps give public companies access to substantial permanent capital, enabling them to undertake complex megaprojects.

Routines to control mistakes and misjudgments also impose unintended but unavoidable restrictions. They discourage small projects whose profit potential cannot absorb the fixed planning and oversight costs. Requirements for objective evidence and consensus similarly deter initiatives with high uncertainty about customers, technologies, and competitors. For example, large corporations avoid highly novel projects, as mentioned, or more generally, initiatives where they cannot objectively evaluate paths to sustainable profits.

The unintended restrictions provide space for self- financed (or informally financed) entrepreneurs to undertake small, simple initiatives in unsettled markets and rely on their personal capacities to seize fleeting opportunities. Likewise, professional venture capitalists (VCs) and angel investors are not as spontaneous or ad- hoc as self- financed entrepreneurs, but their justificatory requirements are not as strict as those of large public corporations. These

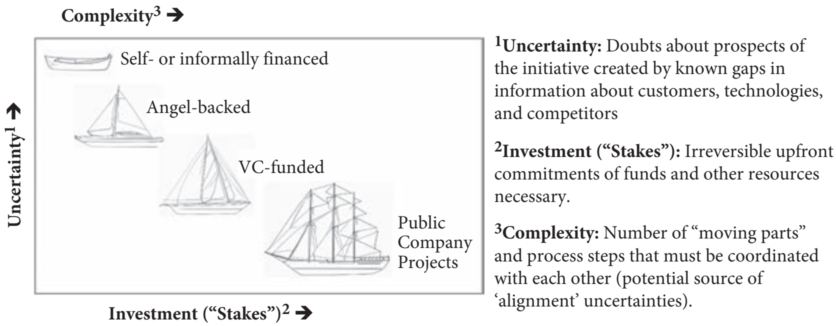

Figure 4.1 Map of Entrepreneurial Specialization[5]

"in-between" requirements encourage VCs and angel investors to specialize in ventures involving intermediate funding needs, uncertainty, and complexity (as shown in Figure 4.1).

This hypothesis about specialization emphasizes the value of a diverse entrepreneurial ecology. For example, venture capital alone cannot sustain widespread opportunities for material advancement and creative adventure. The underlying uncertainty-based reasoning likewise helps us see the advantages of often-mocked large-company routines. These routines do not just help the trains run on time. They undergird the development of railroads and are as indispensable for complex technological advances as time-consuming jury trials are for a civilized criminal justice system.

My hypothesis also suggests that economically significant initiatives cluster along a diagonal running from the northwestern to the southeastern corner of my map. Projects with high uncertainty and high resource requirements (the empty northeastern space in Figure 4.1) are natural nonstarters. Entrepreneurs cannot self-finance such projects, while in-house promoters cannot satisfy the strict routines of large organizations that have the necessary funds and coordination capacities. Conversely, when uncertainty is low (i.e., below the diagonal), so are the profit prospects—and thus also the scope for outside financing.*

Imaginative Discourse Entrepreneurial ideas emerge from a creative process that combines facts and imagination. Entrepreneurs do not merely observe or notice facts. They imaginatively interpret what they observe or notice. Imagination is particularly important in entrepreneurship that goes beyond simply buying

* The southwestern space is not empty, however. Founders of low-potential / low-uncertainty businesses—who just want to find jobs for themselves—cluster there (as discussed in chapter 14).

low to sell high. The innovative entrepreneur must imagine what could be and a plausible path for getting there. Almost by definition (and Knight's thesis), information gaps preclude deducing the desired destination and path through logic or statistical analysis. If sufficient information for logical or reliable statistical inference existed, there would be no opportunity for profit, as I keep reminding students in my entrepreneurship classes.

Moreover, promoters of an enterprise cannot just imagine desired future states and feasible paths. They must also persuade financiers, customers, employees, and others that their imagined scheme is worth supporting. Here, too, promoters cannot merely provide objective facts and signal their confidence in their ventures by putting their own money in them. They must also engage in imaginative discourse that is more writerly than logico-scientific, using figurative language, metaphors, and imagined events.

The discourse entails paradoxes: for example, adding imagined detail makes imagined futures and paths more believable. A detailed spreadsheet model with made-up but defensible numbers is better than not having a model. This is not a con job aimed at naive targets or the peddling of a trippy, hallucinatory fantasy. Imaginative justification and discourse are necessary to overcome the doubts that discourage hard-nosed investors from funding, astute employees from joining, and skeptical customers from buying the offerings of entrepreneurial ventures. Yet the justifications must be *groundedly* imaginative, not pure fantasy.

Broadening Mainstream Views The kind of specialization shown in Figure 4.1—and the routines that help shape it—fall outside the scope of mainstream economic models in which no justification is demanded or supplied. These conventional models emphasize problems of slacking, lying, and cheating (controlled by aligning incentives) rather than misjudgments (controlled by justificatory routines).

Economic theories also typically ignore stories or other kinds of rich communication used to justify or share ideas. In textbook theories of decentralized markets, autonomous buyers and sellers have fully formed preferences and transact if the price is right. Or, as in Hayek's paper mentioned earlier, price changes coordinate dispersed producers and consumers: rising (or falling) prices alone evoke changes in supply and demand. Conversely, theories of hierarchical control posit top-down commands or ratification of subordinates' proposals.[6] In either case, there are no mutual doubts about fallible judgments, no anxiety about outcomes, and no constructive role for storylike discourse. According to these theories, persuasive *pathos* can only deceive.[7] Dialogue to forestall or correct coordination problems is also absent.

Excluding narrative-mode discourse (chapter 18) raises questions about the scope of economics. The Scientific Revolution attacked and replaced animism with naturalistic theories about inert objects that had no will of their own. Scientific economics brought the ethos of the natural sciences into the human sphere, modeling free markets without any actual free will (although it kept the Aristotelian concept of predestined striving). This unnatural naturalism limits our understanding of how we live and work together and our discourse about its improvement. Adding the sensibilities found in the humanities (particularly history and literature) and the law (which stresses human intentions, doubts, and imaginations) thus has both theoretical and practical value.

My book takes a few steps toward such broadening. According to an influential 'paradigmatic' view (reviewed in the next chapter), specific, archetypal problems and solutions shape normal scientific research. Concrete, internalized examples, not abstract constructs, guide what scientists actually do.[8] Similarly, even if mainstream economists question my general conjectures, I hope the concrete applications help broaden their acceptable problems and explanations in areas beyond entrepreneurship.

3. What Lies Ahead

Part 2 (chapters 5–12) describes how mainstream economic theories* obstruct an uncertainty-based view of enterprise, but some forgotten ideas provide helpful guidance. This is the most technical part of the book and the one in which the reactions of economists and noneconomists are most likely to diverge.

I hope economists excuse my simplified comparisons of mainstream and forgotten ideas. My intention isn't to disparage mainstream theories but rather to show how the forgotten theories could improve our understanding of the present-day world. Exploring the unfamiliar may also deepen economists' understandings of the familiar. As Rudyard Kipling's 1891 poem asks, "What do they know of England who only England know?"

Other readers may find the chapters a useful (if idiosyncratic and partial) introduction to important economic ideas, even if they struggle with unfamiliar terms and constructs. Both economists and noneconomists, I hope, find the intellectual histories and biographical sketches included in the chapters

* I rely on representation in National Bureau of Economic Research (NBER) working groups and in articles published in the top financial and economic journals as proxies for "mainstream" ideas. I do not, however, comprehensively review the mainstream specialties. I focus instead on placing my conjectures about uncertainty in mainstream "microeconomic" maps while saying little about macroeconomic aggregates.

stimulating. (Moreover, the modular structure allows readers who find the material gets too simplistic or confusing to skip ahead to Parts 3 and 4).

Part 3 (chapters 13–17) contains the first of the two applications summarized above, on the specialization of entrepreneurship. The chapters proceed down the "southeasterly diagonal" of Figure 4.1: chapter 14 covers self-financed entrepreneurs; chapter 15, ventures backed by angel investors and VCs; and chapters 16 and 17, projects undertaken by large public companies.

Part 4 (chapters 18–21) covers the role of imaginative discourse. The chapters are at once furthest from standard economics, prevailing popular views of storytelling, and my own beliefs before I started this book. The concluding chapter—a Part 5 coda—pleads for widening the field of view of economics, without abandoning existing ideas. It also examines broader social and political problems, beyond commercial enterprise, through the lens of uncertainty.